

ASSIGNMENT

1. Angles of a triangle are in ratio 4 : 1 : 1. Find the ratio between its greatest side and perimeter.

Sol

$$A : B : C = 4 : 1 : 1$$

$$\Rightarrow A = 4k, B = k, C = k$$

$$A + B + C = \pi \quad (\text{ASP of triangle})$$

$$\Rightarrow 4k + k + k = \pi$$

$$\Rightarrow 6k = \pi$$

$$\Rightarrow k = \frac{\pi}{6}$$

$$\therefore A = \frac{2\pi}{3}, B = \frac{\pi}{6}, C = \frac{\pi}{6}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = m$$

$$\Rightarrow \frac{a}{\sin \frac{2\pi}{3}} = \frac{b}{\sin \frac{\pi}{6}} = \frac{c}{\sin \frac{\pi}{6}} = m$$

$$\Rightarrow a = \frac{\sqrt{3}m}{2}, b = \frac{m}{2}, c = \frac{m}{2}$$

$$\text{greatest side} = \left(\frac{\sqrt{3}}{2} \right) m$$

$$\text{perimeter} = a + b + c$$

$$= \frac{\sqrt{3}m}{2} + \frac{m}{2} + \frac{m}{2}$$

$$= \left(\frac{\sqrt{3} + 2}{2} \right) m$$

$$\begin{aligned} \therefore \frac{\text{greatest side}}{\text{perimeter}} &= \frac{\left(\frac{\sqrt{3}}{2} \right) m}{\left(\frac{\sqrt{3} + 2}{2} \right) m} \\ &= \frac{\sqrt{3}}{\sqrt{3} + 2} \end{aligned}$$

2. In a triangle ABC, if $b = 2$, $c = 4$ and $\angle B = \frac{\pi}{3}$ the number of such triangles is _____.
(a) 1 (b) 2 (c) 0 (d) infinite

Sol

$$b = 2, c = 4, B = \frac{\pi}{3}$$

$$\frac{\sin B}{b} = \frac{\sin C}{c}$$

$$\Rightarrow \frac{\sin \frac{\pi}{3}}{2} = \frac{\sin C}{4}$$

$$\Rightarrow \sin C = 2 \sin \frac{\pi}{3} = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

$$\sin C \in [-1, 1] \Rightarrow \text{No such triangle exists}$$

3. Show that, in any ΔABC $a \sin(B - C) + b \sin(C - A) + c \sin(A - B) = 0$.

Sol

$$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow a \sin B = b \sin A$$

Similarly, $b \sin C = c \sin B$ and $c \sin A = a \sin C$

$$\text{LHS} = a \sin(B - C) + b \sin(C - A) + c \sin(A - B)$$

$$= a \sin B \cos C - a \sin C \cos B + b \sin C \cos A - b \sin A \cos C + c \sin A \cos B - c \sin B \cos A$$

$$= b \sin A \cos C - a \sin C \cos B + c \sin B \cos A - b \sin A \cos C + a \sin C \cos B - c \sin B \cos A$$

$$= 0$$

$$= \text{RHS}$$

4. In a ΔABC , prove that $\frac{a-b}{a+b} = \cot\left(\frac{A+B}{2}\right) \tan\left(\frac{A-B}{2}\right)$.

Sol

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k \Rightarrow a = k \sin A, b = k \sin B, c = k \sin C$$

$$\begin{aligned} \text{LHS} &= \frac{a-b}{a+b} \\ &= \frac{k \sin A - k \sin B}{k \sin A + k \sin B} = \frac{\sin A - \sin B}{\sin A + \sin B} \\ &= \frac{2 \cos\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)}{2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)} \\ &= \cot\left(\frac{A+B}{2}\right) \tan\left(\frac{A-B}{2}\right) \\ &= \text{RHS} \end{aligned}$$

5. In a $\triangle ABC$, if $\angle B = 30^\circ$ and $c = \sqrt{3}b$ then $\angle A$ can be _____.

- (a) 45° (b) 60° (c) 90° (d) 120°

Sol

$$\begin{aligned} B &= 30^\circ \\ c &= \sqrt{3}b \\ \frac{a}{\sin A} &= \frac{b}{\sin B} = \frac{c}{\sin C} \\ \Rightarrow \frac{b}{\sin 30^\circ} &= \frac{\sqrt{3}b}{\sin C} \\ \Rightarrow \sin C &= \frac{\sqrt{3}}{2} \Rightarrow C = 60^\circ, 120^\circ \\ \therefore A &= 180^\circ - (30^\circ + 60^\circ) \text{ or } 180^\circ - (30^\circ + 120^\circ) \\ \Rightarrow A &= 90^\circ \text{ or } 30^\circ \end{aligned}$$

6. Prove that $(a^2 - b^2 - c^2) \tan A + (a^2 - b^2 + c^2) \tan B = 0$ in any $\triangle ABC$.

Sol

$$\begin{aligned} \text{LHS} &= (a^2 - b^2 - c^2) \tan A + (a^2 - b^2 + c^2) \tan B \\ &= -2bc \cos A \tan A + 2ac \cos B \tan B \\ &= -2c(b \sin A - a \sin B) \\ &= 0 \\ &= \text{RHS} \end{aligned}$$

7. If $a : b : c = 4 : 5 : 6$ in a triangle ABC with the usual notations, then show that $\angle C = 2\angle A$.

Sol

$$a = 4, b = 5, c = 6 \Rightarrow C > B > A$$

$$\begin{aligned}\cos A &= \frac{b^2 + c^2 - a^2}{2bc} \\ &= \frac{5^2 + 6^2 - 4^2}{2(5)(6)} \\ &= \frac{25 + 36 - 16}{60} \\ &= \frac{45}{60} \\ &= \frac{3}{4}\end{aligned}$$

$$\begin{aligned}\cos B &= \frac{c^2 + a^2 - b^2}{2ca} \\ &= \frac{6^2 + 4^2 - 5^2}{2(6)(4)} \\ &= \frac{36 + 16 - 25}{48} \\ &= \frac{27}{48} \\ &= \frac{9}{16}\end{aligned}$$

$$\begin{aligned}\cos C &= \frac{a^2 + b^2 - c^2}{2ab} \\ &= \frac{4^2 + 5^2 - 6^2}{2(4)(5)} \\ &= \frac{16 + 25 - 36}{40} \\ &= \frac{5}{40} \\ &= \frac{1}{8}\end{aligned}$$

$$\begin{aligned}\cos 2A &= 2\cos^2 A - 1 \\ &= 2\left(\frac{3}{4}\right)^2 - 1 \\ &= 2\left(\frac{9}{16}\right) - 1 \\ &= \frac{9}{8} - 1 \\ &= \frac{1}{8} \\ &= \cos C\end{aligned}$$

$$\therefore C = 2A$$

8. If $\cos \frac{A}{2} = \sqrt{\frac{b+c}{2c}}$ then _____.

- (a) $a^2 + b^2 = c^2$ (b) $b^2 + c^2 = a^2$ (c) $c^2 + a^2 = b^2$ (d) None of these

Sol

$$\begin{aligned}\cos \frac{A}{2} &= \sqrt{\frac{b+c}{2c}} \\ \Rightarrow \cos^2 \frac{A}{2} &= \frac{b+c}{2c} \\ \Rightarrow \frac{1+\cos A}{2} &= \frac{b+c}{2c} \\ \Rightarrow 1 + \frac{b^2+c^2-a^2}{2bc} &= \frac{b+c}{c} \\ \Rightarrow \frac{2bc+b^2+c^2-a^2}{2bc} &= \frac{b+c}{c} \\ \Rightarrow 2bc+b^2+c^2-a^2 &= 2b^2+2bc \\ \Rightarrow a^2+b^2 &= c^2\end{aligned}$$

9. In any ΔABC show that $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2+b^2+c^2}{2abc}$

Sol

$$\begin{aligned}\text{LHS} &= \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} \\ &= \frac{b^2 + c^2 - a^2}{2abc} + \frac{c^2 + a^2 - b^2}{2abc} + \frac{a^2 + b^2 - c^2}{2abc} \\ &= \frac{a^2 + b^2 + c^2}{2abc} \\ &= \text{RHS}\end{aligned}$$

10. In any ΔABC show that $c^2 = (a-b)^2 \cos^2 \frac{C}{2} + (a+b)^2 \sin^2 \frac{C}{2}$

Sol

$$\begin{aligned}\text{RHS} &= (a-b)^2 \cos^2 \frac{C}{2} + (a+b)^2 \sin^2 \frac{C}{2} \\ &= a^2 \cos^2 \frac{C}{2} + b^2 \cos^2 \frac{C}{2} - 2ab \cos^2 \frac{C}{2} + a^2 \sin^2 \frac{C}{2} + b^2 \sin^2 \frac{C}{2} + 2ab \sin^2 \frac{C}{2} \\ &= a^2 + b^2 + 2ab \left(\sin^2 \frac{C}{2} - \cos^2 \frac{C}{2} \right) \\ &= a^2 + b^2 - 2ab \cos C \\ &= c^2 \\ &= \text{LHS}\end{aligned}$$

11. In a ΔABC , show that $(b+c)\cos A + (c+a)\cos B + (a+b)\cos C = a+b+c$.

Sol

$$\begin{aligned}\text{LHS} &= (b+c)\cos A + (c+a)\cos B + (a+b)\cos C \\ &= b\cos A + c\cos A + c\cos B + a\cos B + a\cos C + b\cos C \\ &= b\cos C + c\cos B + c\cos A + a\cos C + b\cos A + a\cos B \\ &= a+b+c \\ &= \text{RHS}\end{aligned}$$

12. In a ΔABC if $c \cos^2 \frac{A}{2} + a \cos^2 \frac{C}{2} = \frac{3b}{2}$ then a, b, c are in _____.

- (a) AP (b) GP (c) HP (d) None of these

Sol

$$\begin{aligned}
 c \cos^2 \frac{A}{2} + a \cos^2 \frac{C}{2} &= \frac{3b}{2} \\
 \Rightarrow c[1 + \cos A] + a[1 + \cos C] &= 3b \\
 \Rightarrow c + c \cos A + a + a \cos C &= 3b \\
 \Rightarrow c + a + b &= 3b \\
 \Rightarrow c + a &= 2b \\
 \Rightarrow a, b, c \text{ are in AP}
 \end{aligned}$$

13. In a ΔABC , show that $(a+b+c)\left(\tan \frac{A}{2} + \tan \frac{B}{2}\right) = 2c \cot \frac{C}{2}$.

Sol

$$\begin{aligned}
 \text{LHS} &= (a+b+c)\left(\tan \frac{A}{2} + \tan \frac{B}{2}\right) \\
 &= 2s \left[\sqrt{\frac{(s-b)(s-c)}{s(s-a)}} + \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} \right] \\
 &= 2\sqrt{s(s-c)} \left[\sqrt{\frac{s-b}{s-a}} + \sqrt{\frac{s-a}{s-b}} \right] \\
 &= 2\sqrt{s(s-c)} \left[\frac{s-b+s-a}{\sqrt{(s-a)(s-b)}} \right] \\
 &= 2c \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} \\
 &= 2c \cot \frac{C}{2} \\
 &= \text{RHS}
 \end{aligned}$$

14. If the sides a, b, c of a ΔABC are in AP then prove that $\cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2}$ are in AP.

Sol

$$\begin{aligned}
 a, b, c \text{ are in AP} \\
 \Rightarrow s-a, s-b, s-c \text{ are in AP} \\
 \Rightarrow \frac{s(s-a)}{\Delta}, \frac{s(s-b)}{\Delta}, \frac{s(s-c)}{\Delta} \text{ are in AP} \\
 \Rightarrow \cot \frac{A}{2}, \cot \frac{B}{2}, \cot \frac{C}{2} \text{ are in AP}
 \end{aligned}$$

15. Prove that in any ΔABC $(abc) \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} = \Delta^2$.

Sol

$$\begin{aligned}\text{LHS} &= (abcs) \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \\ &= (abcs) \sqrt{\frac{(s-b)(s-c)}{bc}} \sqrt{\frac{(s-c)(s-a)}{ca}} \sqrt{\frac{(s-a)(s-b)}{ab}} \\ &= \frac{(abcs)(s-a)(s-b)(s-c)}{abc} \\ &= s(s-a)(s-b)(s-c) \\ &= \Delta^2 \\ &= \text{RHS}\end{aligned}$$